

# Skills Summary

## Place Value to 1,000: Three Ways to Write a Number

*Use this reference while completing your worksheet or when studying.*

Every number below is the same number three times. The place a digit sits in tells you what it is worth: hundreds, tens, ones. Once you know each digit's value, all three forms are just different ways of saying it.

### 1. Standard form to expanded form

**Rule:** give each digit its place value, then join them with +.

- hundreds digit  $\times 100$     tens digit  $\times 10$     ones digit  $\times 1$
- A digit of 0 contributes nothing — leave that part out of the sum.

**Example:** Write 638 in expanded form.

**Step 1.** Expanded form asks for the value of each digit, so read the number one place at a time instead of as a whole.

**Step 2.** 6 is in the hundreds place  $\rightarrow 600$ ; 3 is in the tens place  $\rightarrow 30$ ; 8 is in the ones place  $\rightarrow 8$ .

$$638 = 600 + 30 + 8$$

**Try it:** Write 275 in expanded form.

$$275 = 200 + 70 + 5$$

### 2. Expanded form to standard form

**Rule:** each part of the sum fills exactly one place. Add them up, or just read off the leading digit of each part.

If a place is missing from the sum, that place gets a 0 — the zero holds the place open so the other digits do not slide over.

**Example:** Write  $800 + 50 + 1$  in standard form.

**Step 1.** Each part names one place, so the job is to put one digit in each place rather than to do a long addition.

**Step 2.** 800  $\rightarrow$  8 hundreds; 50  $\rightarrow$  5 tens; 1  $\rightarrow$  1 one.

$$851$$

**Try it:** Write  $400 + 9$  in standard form.

$$409$$

### 3. Number names (word form) to standard form

---

**Rule:** break the name at the word “hundred”.

(*number*) *hundred* → the hundreds digit; the rest of the name is a two-digit number that fills the tens and ones places.

Teen and tens words are *one* two-digit chunk, not two digits written side by side: “fifteen” is 15, which is 1 ten and 5 ones.

**Example:** Write three hundred sixty-two in standard form.

**Step 1.** The name already tells you the place values, so split it at “hundred” and translate each chunk.

**Step 2.** “three hundred” → 300; “sixty” → 60; “two” → 2; so  $300 + 60 + 2$ .

**362**

**Try it:** Write nine hundred four in standard form.

**check:** 904

### 4. Comparing numbers written in different forms

---

**Rule:** put *both* numbers in standard form first, then compare place by place from the left.

Same hundreds? Compare tens. Same tens? Compare ones. Use  $<$  for “is less than” and  $>$  for “is greater than” — the small point aims at the smaller number.

**Example:** Which is greater:  $500 + 20 + 7$  or five hundred seventy-two?

**Step 1.** The two numbers are written in different forms, so nothing can be compared until both are in standard form.

**Step 2.**  $500 + 20 + 7 = 527$  and five hundred seventy-two = 572; both have 5 hundreds, and 2 tens is less than 7 tens.

**$527 < 572$**

**Try it:** Compare six hundred eight and  $600 + 80$ .

**check:**  $608 > 680$

**(!) Watch out:** A zero in the middle is the trap. In 907 the tens place is empty, so the expanded form is  $900 + 7$ , not  $90 + 7$ . And in  $600 + 5$  the missing tens part still needs a 0 in standard form: the answer is 605, never 65.